STANDARD MICROSYSTEMS CORPORATION
35 Marcus Blvd., Hauppauge, N.Y. 11787
(516) 273-3100 · TWX-510-227-8898
We keep ahead of our competition so you can keep ahead of yours.

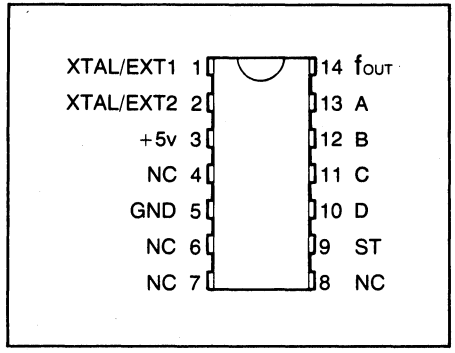
COM 8126
COM 8126T

Baud Rate Generator
Programmable Divider

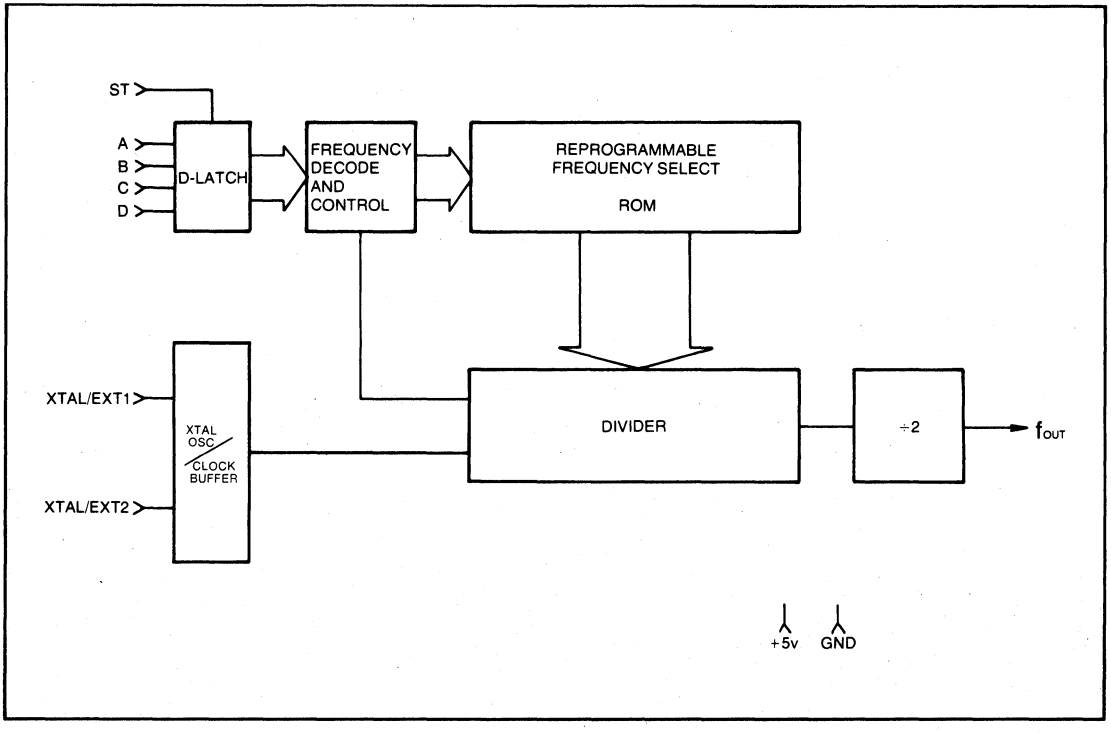
FEATURES

- ☐ On chip crystal oscillator or external frequency input
- ☐ Single +5v power supply
- ☐ Choice of 16 output frequencies
- ☐ 16 asynchronous/synchronous baud rates
- ☐ Direct UART/USRT/ASTRO/USYNRT compatibility
- ☐ Re-programmable ROM via CLASP® technology allows generation of other frequencies
- ☐ TTL, MOS compatibility
- ☐ Compatible with COM 5026

PIN CONFIGURATION



BLOCK DIAGRAM



SECTION VI

General Description

The Standard Microsystem's COM 8126 is an enhanced version of the COM 5026 Baud Rate Generator. It is fabricated using SMC's patented COPLAMOS® and CLASP® technologies and employs depletion mode loads, allowing operation from a single +5v supply.

The standard COM 8126 is specifically dedicated to generating the full spectrum of 16 asynchronous/synchronous data communication frequencies for 16X UART/USRT devices. A large number of the frequencies available are also useful for 1X and 32X ASTRO/USYNRT devices.

The COM 8126 features an internal crystal oscillator which may be used to provide the master reference frequency. Alternatively, an external reference may be supplied by applying complementary TTL level signals to pins 1 and 2. Parts suitable for use only with an external TTL reference are marked COM 8126T. TTL outputs used to drive the COM 8126 or COM 8126T XTAL/EXT inputs should not be used to drive other TTL inputs, as noise immunity may be compromised due to excessive loading.

The output of the oscillator/buffer is applied to the divider for generation of the output frequency. The divider is capable of dividing by any integer from 6 to $2^{19} + 1$, inclusive. If the divisor is even, the output will be square; otherwise the output will be high longer than it is low by one fx clock period.

The divisor ROM contains 16 divisors, each 19 bits wide, and is fabricated using SMC's unique CLASP® technology. This process permits reduction of turnaround time for ROM patterns. The four divisor select bits are held in an externally strobed data latch. The strobe input is level sensitive: while the strobe is high, data is passed directly through to the ROM. Initiation of a new frequency is effected within 3.5μs of a change in any of the four divisor select bits (strobe activity is not required). This feature may be disabled through a CLASP® programming option causing new frequency initiation to be delayed until the end of the current f_{OUT} half-cycle. The divisor select inputs have pull-up resistors; the strobe input does not.

Description of Pin Functions

Pin No.	Symbol	Name	Function
1	XTAL/EXT1	Crystal or External Input 1	This input is either one pin of the crystal package or one polarity of the external input.
2	XTAL/EXT2	Crystal or External Input 2	This input is either the other pin of the crystal package or the other polarity of the external input.
3	V_{CC}	Power Supply	+5 volt supply
4,6,7,8	NC	No Connection	
5	GND	Ground	Ground
9	ST	Strobe	A high level strobe loads the input data (A, B, C, D) into the input divisor select register. This input may be strobed or hard-wired to a high level.
10-13	D,C, B, A	Divisor Select Data Bits	The logic level on these inputs as shown in Table 1, selects the output frequency.
14	f_{OUT}	Output Frequency	This output runs at a frequency selected by the divisor select data bits.

For electrical characteristics, see page 199.